

Senior High School



Learning Activity Sheet for General Mathematics

Quarter 3

Unit

9

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LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	3
Lesson No.:	9.1	Date:	
Lesson Title/Topic:	Discrete Random Variables		
Name:		Grade & Section:	

I. Activity No. 1	Practicing Learned Skills																																		
II. Objective:	At the end of the activity, you be able to apply your understanding of random variables in real-life situations																																		
III. Materials Needed:	pen calculator																																		
IV. Instructions:	Instructions: Perform the indicated tasks. Write your answer on a sheet paper. A. Identifying Discrete or Continuous Random Variables - Identify whether each random variable is Discrete (D) or Continuous (C). - Number of learners absent in a class - Time (in minutes) spent answering a test - Number of cars passing a toll booth in an hour - Weight of a newborn baby - Number of defective items in a box B. Construct probability distribution table for each probability experiment and draw the graph. - Number of heads in tossing a coin twice. - Getting odd numbers in rolling a die. - Number of tails in tossing three coins. - Getting a heart when drawing a card from standard deck. - Sum of numbers on the two dice. C. Find the mean, variance, and standard deviation of each problem, then interpret the results. - A learner buys snacks during recess. The number of snacks she buys per day (X) follows this distribution:<table><tr><td>Number of snacks (X)</td><td>0</td><td>1</td><td>2</td></tr><tr><td>P(X)</td><td>0.2</td><td>0.5</td><td>0.3</td></tr></table> - A tricycle driver observes the number of passengers picked up in one trip.<table><tr><td>Passengers (X)</td><td>1</td><td>3</td><td>5</td></tr><tr><td>P(X)</td><td>0.4</td><td>0.4</td><td>0.2</td></tr></table> - A customer service agent records the number of calls received in one hour.<table><tr><td>Calls (X)</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>P(X)</td><td>0.1</td><td>0.3</td><td>0.4</td><td>0.2</td></tr></table> typically handles about 2 calls per hour, with relatively low fluctuation. - The number of attempts a learner needs to pass an online quiz is recorded.<table><tr><td>Attempts (X)</td><td>2</td><td>4</td><td>6</td></tr><tr><td>P(X)</td><td>0.5</td><td>0.3</td><td>0.2</td></tr></table> - A barangay health worker records the number of family members who received a flu vaccine in a day. The probability distribution is:	Number of snacks (X)	0	1	2	P(X)	0.2	0.5	0.3	Passengers (X)	1	3	5	P(X)	0.4	0.4	0.2	Calls (X)	0	1	2	3	P(X)	0.1	0.3	0.4	0.2	Attempts (X)	2	4	6	P(X)	0.5	0.3	0.2
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		Family Members Vaccinated (X)	0	1	2	3
		P(X)	0.1	0.4	0.3	0.2
V. Synthesis	Answer the following questions: 1. How did identifying the random variable and its type help you understand the problem better? 2. What does the mean, variance, and standard deviation tell you about the expected outcomes and variability in each scenario? 3. How can understanding random variables and their measures help you make better decisions in real-life situations involving uncertainty?					

I. Activity No. 2	Making Real-World Connections										
II. Objective:	At the end of the activity, you be able to apply your understanding of random variables in real-life situations.										
III. Materials Needed:	tape measure or meter stick pen and paper calculator										
IV. Instructions:	Activity C.1: Making Real-World Connections Instructions: Analyze the scenario and perform the indicated task. Scenario: Your school canteen wants to improve its service by stocking the snacks and drinks that learner like the most. A survey will be conducted to find out which snack and drink combination learners prefer. The combinations to be surveyed are: <table border="1"> <thead> <tr> <th>Snack</th><th>Drink</th></tr> </thead> <tbody> <tr> <td>Chips</td><td>Juice</td></tr> <tr> <td>Sandwiches</td><td>Water</td></tr> <tr> <td>Fruits</td><td>Milk</td></tr> <tr> <td>Cookies</td><td>Soda</td></tr> </tbody> </table> The random variable (X) in this investigation is the number of learners who prefer each snack and drink combination. This is a discrete random variable because it can be counted. You will conduct survey to a sample of 20–30 classmates, record their preferred snack and drink combination, and organize the results in a table. Then, they will calculate the expected value (mean), variance, and standard deviation of the number of learners preferring each combination to determine which combination is most popular and how much variation exists among learners' preferences. Finally, you will interpret the results and make recommendations to the canteen about which snack and drink combinations to stock more or less based on the data collected.	Snack	Drink	Chips	Juice	Sandwiches	Water	Fruits	Milk	Cookies	Soda
Snack	Drink										
Chips	Juice										
Sandwiches	Water										
Fruits	Milk										
Cookies	Soda										
V. Synthesis:	Response to each question. • How did calculating the mean, variance, and standard deviation help you understand learners' snack and drink preferences?										

	• Which snack and drink combination was most preferred, and how does the variability affect your recommendation? • How can understanding random variables and their measures help you make better decisions in real-life situations?
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I. Activity No. 3	Assessing Learning Outcomes
II. Objective:	At the end of the activity, you be able to apply your understanding of random variables in real-life situations
III. Materials Needed:	pen paper calculator
IV. Instructions:	Read and answer each item. - Which of the following best defines estimation? A. Converting one unit to another B. Measuring an object using a ruler C. Calculating the exact length of an object D. Making an approximate judgment of a value or quantity - If one meter is approximately the distance from your shoulder to your fingertips, what does this show? A. The use of a trundle wheel B. A standard measuring tool C. The length of a small object D. An example of estimation using a familiar reference - Which of the following is the best tool to measure the length of a football field? A. Ruler B. Tape measure C. Trundle wheel D. Odometer - A ribbon costs ₱20 per meter. If you need 4.5 meters, how much will you pay? A. ₱80 B. ₱85 C. ₱90 D. ₱100 - An odometer reading increased from 25,320 km to 25,645 km after a trip. How far did the car travel? A. 225 km B. 315 km C. 325 km D. 335 km - Which of the following situations best shows when estimation is useful? A. Exact measurement is always required B. We want to save time and get a quick idea of size or distance C. We are building machines that require precision D. We cannot use measuring tools at all - A dressmaker needs 2.5 meters of fabric for one dress. If she makes 3 dresses, how many meters of fabric are needed in total? A. 5.5 m B. 6.5 m C. 7.5 m D. 8.5 m - Why might two learners measuring the same desk using hand spans get different results? A. The desk changed size B. They used different rulers C. Their hand spans are not of equal length D. One student miscounted the desk's corners - Which statement best explains the advantage of using standard measuring tools over body parts? A. They are faster to use.

	B. They are less expensive. C. They are smaller and lighter. D. They are more accurate and consistent. 10. You are asked to measure the length of a school playground but do not have a measuring tape. What is the best strategy to obtain an approximate measurement? A. Guess the length by looking at it B. Wait until someone brings a ruler C. Use your notebook to measure repeatedly D. Use large steps or a trundle wheel if available to estimate the distance.
V. Synthesis:	Answer the following questions: 1. Have you completed all the tasks correctly? 2. Which items did you find easy, and which did you find challenging? 3. What common errors did you make? 4. How will you address your misconceptions?

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	3
Lesson No.:	9.2	Date:	
Lesson Title/Topic:	Normal Distribution		
Name:		Grade & Section:	

I. Activity No. 1	Practicing Learned Skills
II. Objectives:	At the end of the activity, you should be able to apply your knowledge and skills on normal distribution into real life scenario,
III. Materials Needed:	pen paper calculator
IV. Instructions:	Instructions: Do as indicated. Write your answer on a separate sheet of paper. Part A. 1. On graph paper, sketch a bell curve for test scores with mean = 50 and standard deviation = 10. 2. Mark the intervals: 40–60, 30–70, and 20–80. 3. Shade each region and label the approximate % of data it covers. Part B. 1. Compute z for these cases: a. $X = 70, \mu = 60, \sigma = 5$ b. $X = 45, \mu = 50, \sigma = 10$ 2. Convert back to raw scores X given: a. $z = 2, \mu = 100, \sigma = 15$ b. $z = -1.2, \mu = 80, \sigma = 8$ Part C. Read the given scenario: IQ scores are normally distributed with mean $\mu = 100$ and standard deviation $\sigma = 15$. Find the following probabilities: a. $P(X < 115)$ b. $P(X > 85)$ c. $P(85 < X < 115)$ Part D. Read the given scenario: SAT scores are normally distributed with mean $\mu = 500$ and standard deviation $\sigma = 100$. Answer the following questions: a. What percentage of learners score above 650? b. What score marks the lowest 25% of learners?
V. Synthesis	Answer the question briefly. - What formula is used to convert a raw score (X) into a z-score? - If $X = 650$, what z-score will this correspond to? - What does the z-score tell us about the position of 650 relative to the mean?

	4. When solving for probabilities greater than a score (e.g., $P(X > 650)$), do we directly use the z-table value or subtract from 1? Why? 5. What does the phrase “lowest 25% of learners” mean in terms of area under the normal curve? 6. From the z-table, what approximate z-score corresponds to the 25th percentile? 7. How do you convert a z-score back into a raw score (X)?
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I. Activity No. 2	Making Real-World Connection
II. Objectives:	At the end of the activity, you should be able to apply your knowledge and skills on normal distribution into real life scenario,
III. Materials Needed:	pen paper calculator
IV. Instructions:	Part A. Data Collection - Conduct a quick survey using the question: “How many hours do you spend on screen (phone, tablet, laptop) per day?” - Select at least of your classmates or nearby sections. - Record the data in a table. Use whole numbers or nearest 0.5 hour for accuracy. Part B. Data Analysis - Compute the: - Mean (μ) - Standard deviation (σ) - Make an assumption: “We assume that the daily screen time of students follows a normal distribution.” Part C. Application of Normal Distribution - Conversion to Standard Normal Variable Choose one data value from your survey and compute its Z-score using $Z = \frac{X - \mu}{\sigma}$ - Probabilities and Percentiles Using a Z-table or calculator/app: - Find the probability that a randomly selected student spends more than X hours on screen (choose X from your data). - Determine the 75th percentile of screen time. - Problem Solving & Decision-Making Suppose the school plans a digital wellness program for students in the top 25% of screen time users. Determine the screen-time cutoff value. Explain how you obtained this value. Part D. Presentation of Findings - Draw a normal distribution curve. - Label the mean and standard deviation - Shade the region representing the top 25% Write a 2–3 sentence conclusion explaining how the results can guide school decisions.
V. Synthesis	Answer the following questions:

	1. What did you learn about normal distribution after collecting and analyzing real data from your classmates? 2. How did converting raw data to Z-scores and using probabilities or percentiles help you better understand the spread and pattern of the data? 3. Why is normal distribution important in making real-life decisions, such as identifying students who may need support or interventions? 4. What part of today's activity helped you appreciate statistics more, and why?
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I. Activity No. 3	Assessing Learning Outcomes
II. Objectives:	At the end of the activity, you should be able to apply your knowledge and skills on normal distribution into real life scenario,
III. Materials Needed:	pen paper calculator
IV. Instructions:	Read each scenario carefully and perform the task as directed. 1. A test has $\mu = 50$ and $\sigma = 8$. If a learner scores 66, what is the z-score? 2. If $z = -1.25$, $\mu = 100$, $\sigma = 12$, what is the raw score? 3. The scores on a standardized test are normally distributed with a mean of 500 and a standard deviation of 90. Determine: a. $P(X < 560)$ b. $P(X > 440)$ 4. The weights of bags of rice are normally distributed with $\mu = 50$ kg and $\sigma = 2$ kg. a. What % of the bags weigh more than 53 kg? b. A retailer only accepts bags in the middle 80% of weights. What is the acceptable weight range?
V. Synthesis	Answer the question briefly. 1. Which question did you find the most challenging, and why? 2. How did calculating Z-scores or probabilities help you understand the scenario better? 3. Can you think of a real-life situation where you might use what you learned in this activity?

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	3
Lesson No.:	9.3	Date:	
Lesson Title/Topic:	Sampling		
Name:		Grade & Section:	

I. Activity No. 1	Practicing Learned Skills
II. Objective:	At the end of the activity, you should be able to apply the concepts of sampling into real life scenario.
III. Materials Needed:	pen paper
IV. Instructions:	Read each activity carefully and perform the task as directed. Option 1: Classroom Simulation Part I. Fill-in-the-Blank Sampling! Instructions: Complete each sentence. 1. A _____ is the entire group being studied 2. A _____ is a smaller part of the population 4. A _____ is a characteristic of a sample. (Statistic) 5. _____ means everyone has an equal chance of being chosen. Part II. Show Us What You Got! Instructions: 1. Each group will act out one sampling method using classmates or name cards. 2. Decide if it is probability or nonprobability sampling. 3. Explain why sample size and method matter. Does your method give accurate results? Is it quick and easy, or fair and representative? Part III. Sample It Right! Instructions: 1. Each group will be provided with scenario cards describing different sampling situations a. A teacher wants to know the preferred mobile phone brand of all Grade 11 learners in your school. Every learner should have an equal chance of being selected. b. The librarian wants to survey learners about their reading habits. She lines up all the learners who enter the library in one week and selects every 5th learner to answer the survey. c. The learner council wants to survey 100 learners about their study habits. Since the school has three strands (STEM, HUMSS, GAS), they want to make sure each strand is fairly represented in the sample. d. A barangay health worker wants to survey families about waste management. Instead of selecting households from all over the barangay, she randomly chooses two whole streets and surveys every household in them. e. A learner researcher wants quick feedback about the cafeteria food. He surveys only the learners sitting at the nearest tables during lunchtime.

- f. A guidance counselor wants to understand the experiences of learners with anxiety. She interviews one learner, who then refers her to another learner with the same concern, and the process continues.
 - g. A science teacher wants to study how top-performing learners prepare for exams. She chooses only the honor learners from each section to participate in the survey.
 - h. A researcher wants to survey 60 learners about their social media habits. To ensure balance, she decides to collect responses from exactly 20 STEM, 20 HUMSS, and 20 GAS learners, regardless of how many are available.
2. The group will identify the population, choose an appropriate sampling method, and decide on sample size.
 3. Your group will present the sampling strategies and justify your choices.

Option 2: **Sampling Challenge**

Part I. Matching Race

Instructions:

1. Work with a partner.
2. You will receive two sets of cards: terms and definitions.

Terms	Definitions
○ Parameter ○ Population ○ Random Sampling ○ Sample ○ Statistic	○ The entire group being studied ○ A smaller part of the population ○ A characteristic of the population ○ A characteristic of the sample ○ Everyone has an equal chance of being chosen

3. Your task is to match each term with its correct definition
4. You have 2 minutes—work quickly, it's a race!
5. The class will check answers together. Fastest correct group earns points or praise.

Part II. **Case Challenge**

Instructions:

1. Read each case scenario carefully.
2. Identify which sampling method is being used.
3. Work in pairs or small groups to discuss your answers.

Case Scenarios:

1. A teacher wants to know the learning style of all A learner quickly surveys classmates in the canteen.
2. Grade 11 learners.
A clinic wants to survey learners with asthma.
3. A researcher wants fair representation from all Track (Academic, TechPro).
4. A barangay officer surveys whole blocks of houses.

Part III. **Sampling Showdown**

Instructions:

1. Look back at the five case scenarios from Part II.
2. Decide which ones are probability sampling and which are nonprobability sampling.
Share your answers with your partner.

	3. Be ready to explain your choices during the class discussion. 4. • Option 3: Sampling Face-Off! Part I. Hot Seat Game Instructions: - One chair in front is the “hot seat.” - Volunteers (or chosen learners) will sit in the hot seat one at a time. - The teacher will ask a question. If you are in the hot seat, you must answer within 10 seconds. - If correct, you earn a point for your group. If not, the chance passes to another learner. Hot Seat Questions - What is a parameter? - What is a statistic? - What is random sampling? - What is the difference between population and sample? - Why is random sampling important? Part II. Face-Off Instructions: - The class will be divided into two teams: - Team A - Argues for Probability Sampling - Team B - Argues for Nonprobability Sampling - Each team will have 2 minutes to prepare arguments. - Debate Rules: - Opening Arguments: Each team speaks for 3 minutes. - Rebuttals: Each team has 2 minutes to respond to the other side. - Q&A: Teams may ask each other questions for 2 minutes. - Be respectful, listen carefully, and support your answers with reasons or examples.
V. Synthesis	Answer the following questions: For Option 1 - What is the target population? - Which sampling method fits this scenario best? - How will you ensure the sample is representative? - How large should the sample be for reliable results? - What challenges might arise in your sampling method? For Option 2. - Which sampling method do you think gives the most reliable results, and why? - Which method would you personally use if you were asked to survey all Grade 11 learners in your school? For Option 3. - Why is choosing the right sample size and method important? - Which sampling method would you choose for school-based research? Why?

I. Activity No. 2	Making Real-World Connections
II. Objective:	At the end of the activity, you should be able to apply the concepts of sampling into real life scenario.
III. Materials Needed:	• pen • paper
IV. Instructions:	Read carefully the scenario and do the tasks as indicated. Scenario: Your school is preparing for a Cultural Celebration Day. The goal is to make sure all tracks feel included the Academic and TechPro. The organizers need to collect learner opinions using a sampling plan that is fair and representative of the whole school. Case 1. Talent Show Line-up The school wants to know what kind of performances learners prefer for the Talent Show. If the survey only asks learners from the Academic track, the results may not reflect the interests of TechPro. Create a sampling plan that represents all tracks fairly. Case 2. Limited Time The event is next week, and the organizers don't have time to survey all 600 learners. They decide to pick sections to answer the survey. If they only choose sections from one track, the results won't be fair. Decide which sections to include so that the survey results represent all tracks. Case 3: Food Fair During the celebration, there will be a Food Fair. Learners from different tracks may have different food preferences. If the survey is done only in one track, the results will not capture everyone's opinion. Make a sampling plan that ensures all track's choices are included. Task: Choose only one the cases and design a sampling plan to collect information from a portion of the learner population. This plan should be fair, unbiased, and representative of the school community. Work together to figure out: • Who to ask? • How to pick them • How many to ask • How to make sure your survey is fair and accurate Answer the questions below. Work quickly but think carefully. Be ready to share your ideas with the class Identify the Population Who is your target population? Who should be included in your survey? Explain why you chose this group. Choose a Sampling Method What method will you use to choose your sample? Why did you choose this method? Estimate the Sample Size

	Suppose your school has 600 learners. To get information, you want to survey enough people. How many learners will you survey? Explain your answer. Consider Fairness Suppose your school has 600 learners. To get information, you want to survey enough people. • What problems or bias might happen in your sampling plan? • How can you reduce or avoid these problems? Twist The school has: 300 boys, 300 girls Academic = 450 TechPro = 150 How would this information affect your sampling method and sample size? 3. Be ready to explain your choices during the class discussion. 4. Reflect: • Which sampling method do you think gives the most reliable results, and why? • Which method would you personally use if you were asked to survey all Grade 11 learners in your school?
V. Synthesis	Answer each question. 1. How did your group decide on the sampling method for your scenario? 2. What made your sample fair and representative of the whole school? 3. Why is it useful to use sampling instead of asking everyone in real-life situations? 4. How could your sampling plan affect the accuracy of the results

I. Activity No. 3	Assessing Learning Outcomes
II. Objective:	At the end of the activity, you should be able to apply the concepts of sampling into real life scenario
III. Materials Needed:	pen paper
IV. Instructions:	Read each direction carefully and perform the tasks as indicated. Part I. Multiple Choices Write the letter of the best answer. Scenario for items 1 and 2: The Learner Council wants to know learners' favorite lunch meal to improve the canteen menu. There are 900 learners in the school. 1. To make the results fair, which method should the Learner Council use? a. Ask only the council officers b. Ask friends during lunch c. Use random sampling so every learner has an equal chance d. Post a survey only on social media 2. If they decide to survey 10% of the learners, how many should they include in the sample? a. 45 b. 90 c. 100 d. 120

	Scenario 2 for items 3 and 4. The principal wants data about all Grade 11 learners' favorite sports. A teacher collects data from one section only and claims it represents all Grade 11 learners. 3. The teacher's result is an example of: - Parameter - Statistic - Population - Sample size 4. Why might the teacher's result not represent all Grade 11 learners? - Because it is too large - Because only one section was included - Because it uses random sampling - Because it surveyed the whole population Scenario for item 5. A barangay health worker wants to survey high school learners about their sleeping habits. She considers two options: Option A (Probability): Randomly selects 100 learners from the master list. Option B (Nonprobability): Asks 100 learners near the covered court after class. 5. Which option is more likely to represent the entire learner population? - Option A, because every learner had an equal chance of being chosen - Option A, because it only uses volunteers - Option B, because it is easier and faster - Option B, because it includes only active learners Part II. Open-Ended (Application & Reasoning) Answer briefly and clearly in 2–3 sentences. - In the school canteen survey, why is random sampling important for fairness? - In the sports festival example, explain the difference between a parameter and a statistic. Give one example of each. - In the community health project, when might the health worker prefer nonprobability sampling (Option B) even if it is less accurate? - Compare probability and nonprobability sampling. Which one would you use for research in your school? Explain why.
V. Synthesis	Response to the question below. - Which question did you find the most challenging, and why? - How did the lesson on sampling help you understand the scenario better? - Can you think of a real-life situation where you might use what you learned in this activity?

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	3
Lesson No.:	Unit 9	Date:	
Lesson Title/Topic:	Performance Task		
Name:		Grade & Section:	

I. Activity No.	Data-Driven Decisions: A Mini Statistical Investigation
II. Objective:	At the end of the activity, you should be able to apply the concepts of sampling into real life scenario
III. Materials Needed:	Pen paper
IV. Instructions:	Goal You will design and carry out a mini statistical investigation about how many hours senior high school learners spend on social media each day, analyze the results using statistical methods (sampling, expected value, variance, standard deviation, normal distribution, z-scores), and produce a clear report and presentation with conclusions and recommendations. Role You are a member of the Learner Research Team responsible for planning, collecting, analyzing, and reporting on a real-life school issue (social media usage). Audience Your report and presentation will be delivered to: Your classmates and teacher (primary audience) School monitoring team or guidance counselor (secondary audience — optional) Situation Senior high students and school staff are concerned about social media use and its effects on study time and well-being. The school needs accurate, data-driven information to inform possible interventions. Your team must produce a reliable mini-study that follows good sampling practice and sound statistical analysis. Product/Performance Submit the following a proposal paper indicated the following information: 1. Sampling Plan (document) — research topic, population, sample size and justification, sampling method and short rationale, parameter vs statistic. 2. Data Collection Sheet — raw data (anonymized). 3. Discrete Data Analysis (report + frequency table) — frequency distribution, expected value (mean), variance, standard deviation, interpretation. 4. Continuous Data Analysis (report + graph) — normal curve sketch (or computer plot), z-score calculations for two chosen values, probabilities/percentiles from z-table, interpretation. 5. Conclusion & Reflection (3–5 sentences) — answer the three reflection prompts. 6. Presentation (3–5 minutes) — a concise slide or poster summarizing methods, key results (numbers & graph), conclusion & recommendation. 7. Include an appendix with the sampling instrument (questionnaire/consent note) and any calculation.

Standards & Success Criteria (Rubric / How performance will be judged)
Below are clear success criteria mapped to the rubric percentages. Use this as a checklist while you work.

Success Criteria (Checklist)

1. Topic, population, and sample size are clearly stated.
2. Sample size is justified (basic reasoning or formula).
3. Appropriate sampling method chosen and briefly explained.
4. Parameter and statistic are correctly identified for your study.
5. Discrete variable and continuous variable are correctly classified and justified.
6. Frequency table is complete and correct.
7. Mean (expected value), variance, and standard deviation are computed correctly.
8. Normal curve is sketched or plotted and labeled (mean, SD, chosen values).
9. Two z-scores correctly computed; corresponding probabilities/percentiles correctly read.
10. Interpretations are clear and link results to the research question.
11. Conclusion answers what the data showed, evaluates sampling method, and explains how statistics supported the conclusion.
12. Presentation is clear, neat, and within time limit.

Rubrics for Performance Task

Criteria	Excellent	Good	Fair	Needs Improvement
Sampling Plan 20%	Clear topic, population, sample size, method, explanation	Mostly clear; minor gaps	Incomplete or weak	Wrong/missing
Variable Classification 15%	Correct discrete & continuous with reasons	Correct but brief	Partly correct	Wrong/missing
Discrete Data Analysis 30%	Correct table, mean, variance, SD, clear interpretation	Minor errors	Several errors/unclear	Wrong/missing
Normal Distribution 25%	Correct z-scores, probabilities, curve, interpretation	Minor errors	Some correct, incomplete	Wrong/missing
Conclusion & Reflection 10%	Clear, thoughtful reflection	Adequate	Minimal/vague	Missing
Presentation 10%	Accurate, neat, organized	Mostly accurate; minor issues	Some errors; messy	Many errors

V. Synthesis

Answer each question.

1. What did you learn about the social media habits of senior high school learners from the data you collected and analyzed?
2. What challenges did you encounter during data collection or analysis, and how did you solve them?
3. Why is it important to use statistics (such as mean, standard deviation, and z-scores) instead of opinions when making conclusions about social media use?